

UNIVERSITY OF CAMERINO

Master of Science in Computer Science

School of Science and Technology

Sleep Apnea Detection

Audio Classification Enhanced by Auxiliary Physiological Supervision

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Chapter 1

Introduction

Sleep apnea is a common chronic sleep disorder that affects both the quality of life of people suffering from the condition and the risk of developing severe health issues, such as cardiovascular diseases. A literature-based analysis estimated that Obstructive Sleep Apnea, the most common type, affects approximately 425 million adults aged 30–69 years worldwide, with at least moderate severity [1]. Despite its high prevalence, the condition remains substantially underdiagnosed. A contributing factor is the burden of performing polysomnography, which is the leading diagnostic procedure and requires the patient to sleep a full night connected to multiple sensors. Less invasive and cheaper approaches would have a huge impact on affected individuals, lowering the barrier to receiving a proper diagnosis.

1.1 Sleep Apnea description

The term *sleep apnea* is an umbrella expression that covers several disorders which share periods of reduced or interrupted airflow during the night, while the underlying mechanisms and specific manifestations may vary. Regarding the severity of the obstruction, a medical diagnosis makes the following distinction:

- **Apnea**: The airflow amplitude must decrease by at least 90% for at least 10 seconds.
- **Hypopnea**: The airflow amplitude must decrease by at least 30% for at least 10 seconds, associated with a decrease of at least 3% in oxygen saturation or an EEG arousal.

Using a polysomnography machine, apneas are primarily identified using an oronasal thermal airflow sensor, while hypopneas are primarily identified using nasal pressure. An accurate reconstruction of these channels may be beneficial to any deep learning model, a hypothesis tested in this project.
Depending on the underlying mechanism, three types of apneas cover most of the events:

- **Obstructive Sleep Apnea** (OSA): the soft tissue in the upper airway relaxes, obstructing the passage of air, even though the respiratory muscles continue to contract.
- **Central Sleep Apnea** (CSA): the respiratory drive from the brain stops, and any muscular effort likewise ceases for the duration of the event.
- **Mixed apnea**: the event begins with a lack of drive (central component), but when the effort resumes, the airway remains obstructed.

The same patient can suffer from a combination of these events in one night.
The principal quantity of interest is the **Apnea–Hypopnea Index** (AHI), which is computed as:

$$\mathrm{AHI} = \frac{N_{\mathrm{apneas}} + N_{\mathrm{hypopneas}}}{T_{\mathrm{sleep}}\ (\mathrm{hours})}$$

Of course, two patients may have the same associated AHI, yet experience a different level of physiological burden, depending on the duration of breathing interruptions, oxygen desaturation, and other factors. Yet, clinically, a diagnosis is mainly based on reported symptoms and the frequency and severity of the events. For this reason, it would be valuable to obtain a precise measurement of the above quantities without invasively attaching the patient to a machine for a full night.

1.2 Methodology overview

Deep learning approaches that work with different combinations of partial PSG signals have been found in the literature. The less invasive ones make use of an ambient microphone placed in the same room as the patient. From audio alone, a model learns to identify both the rhythm of breathing and prolonged pauses, and hopefully find patterns in the sounds associated with snoring, which are quite common in affected individuals.
In this project, I attempt to classify short windows of ambient sound as containing signs of apnea or not. I based my work on Professor Quadrini's 2024 paper [2], especially regarding the visual approach of using 2D CNNs on mel spectrograms. I made different preprocessing decisions, such as introducing the ever more common PCEN normalization (in place of spectral gating). Nevertheless, the main distinctive idea is to use additional physiological signals as a supervisory aid during the pretraining phase as a regression objective, with the hope of building more robust and general features. This hypothesis is tested with a simple ablation, and the methodology appears sound. The physiological signals are only needed during pretraining; at inference, the model uses audio alone.

1.3 AI disclosure

ChatGPT has been extensively used for literature review, code development, and to quickly perform early experiments. In particular, it made it easy to readily test emerging ideas. The main results can be reproduced using the accompanying Jupyter notebooks, which are curated by the author. The report was primarily written by the author.

Chapter 2

The dataset

Both preliminary guiding experiments and the validation of the methodology are performed on the **PSG-Audio** dataset [3]. To avoid overfitting the procedure to repeatedly observed data, only a small subset of patients was ever seen during the exploratory phase.
The dataset contains breathing sounds recorded simultaneously with the streams from a polysomnography machine. The third version was published in 2022 and provides **287 nights** from different patients, totaling **1400 hours** of audio and annotations for **71.000 events**, counting both apneas and hypopneas. While the amount of data might seem staggering, the real difficulty of this project is training a model that can ignore individual patients' nuances and perform well in different environmental conditions. This pertains to the machine learning problem of domain generalization. In this peculiar medical context, domains are distinguished by specific environmental noises, the characteristics of the microphones, geometric aspects such as the distance between the recorder and the patient, differences in breathing patterns, and so on. A model might perform well only on domains that it has been exposed to, by recognizing features of already seen patients. That is why the split between the training, validation, and test sets must be domain-wise, ensuring no repetition of patients among the three.
The full-night recording of each patient is split into segments spanning 1 hour. The raw streams of one segment are collectively stored in an EDF file (*European Data Format*), which is commonly used for medical data. EDF files pertaining to the same patient come paired with an RML file, containing human-made annotations describing the beginning and termination of all respiratory events.

2.1 The data streams

The dataset contains many different acquisition channels, and I will describe the ones that are essential for understanding my specific implementation.

- **Ambient microphone**: a $48\,\mathrm{kHz}$ acquisition device that is placed approximately 1 meter above the patient's head. This microphone is independent of the PSG system and has its own clock, meaning the authors had to synchronize the sound afterward using another microphone connected to the main machine. They applied a zero padding policy on the final missing segment created by the procedure. I found this synchronization effort to be unfit for my purposes, and I decided to better align the audio in a similar but more precise fashion, as explained in subsection 3.1.1.

- **Tracheal microphone**: a $48\,\mathrm{kHz}$ acquisition device that is in contact with the patient's throat. Like the previous microphone, it is independent of the PSG system. The recording of this device is the one used for synchronization purposes.
- **Nasal pressure**: a pressure-sensitive nasal cannula, sampled at $100\,\mathrm{Hz}$. Pressure variations provide an indirect, nonlinear estimate of nasal airflow. When the airway is obstructed, the amplitude of the signal decreases considerably. Of course, this channel loses information about the airflow through the mouth. In the preprocessing chapter, I observe that this channel can be found in two different polarity regimes, which I decide to normalize.
- **Oronasal thermistor**: a temperature-sensitive sensor, sampled at $100\,\mathrm{Hz}$, that works by detecting differences in temperature between inspired and expired air. Unlike the nasal cannula, it can detect airflow through both the nose and mouth. This response is also nonlinear.
- **Thoracic effort**: a respiratory belt placed around the chest, sampled at $100\,\mathrm{Hz}$, that records changes in thoracic expansion and contraction during breathing. The persistence of muscular effort remains the discriminating factor between central and obstructive apneas.
- **Abdominal effort**: a second respiratory belt, placed around the abdomen and sampled at $100\,\mathrm{Hz}$, that measures abdominal movements. Together with the thoracic belt, it provides a richer description of respiratory effort. In particular, the relative timing or asynchrony of the two signals can reveal an obstruction of the upper airways.

2.2 Annotations

An RML file uses an XML-based format and is used to register the annotations produced by the medical staff during the corresponding polysomnographic study. Scored events are characterized by their type, starting time, and duration. Importantly, the annotations were obtained from the physiological signals of the polysomnography, without using the audio recordings.
The RML files mainly contain annotations associated with apnea events (OSA, CSA, or mixed), but they also include sleep-stage annotations, oxygen desaturation episodes, arousals, snoring, and other physiological events.

2.3 Population statistical features

The dataset is strongly imbalanced with respect to apnea severity. In the original 2021 release, 88.7% of the 212 patients were classified as having severe sleep apnea, whereas only 1.4% belonged to the normal category. This distribution is caused by the clinical setting of the data collection procedure, as patients undergoing polysomnography were generally referred for suspected sleep disorders.
It is true that patients with severe sleep apnea exhibit periods of normal breathing, allowing both positive and negative examples to be extracted from the same patient. Nevertheless, a model trained on this dataset is expected to suffer from selection bias, as the breathing of affected individuals might differ substantially from that of the general population.

Chapter 3

Preprocessing

To make an honest comparison with previous work, I train different models using the Ambient microphone and the Tracheal microphone respectively. The different results I obtain with these two devices highlight the need for high quality recordings. I nevertheless insisted on using the Ambient microphone because it is the less invasive of the two recording approaches. Regarding data preprocessing, I applied transformations both to the audio input and to the four physiological PSG channels.

3.1 Audio Preprocessing

3.1.1 Clock synchronization

In this work, physiological signals are used to provide more fine-grained supervision over the audio. Perfectly aligning the latter to the PSG streams is clearly important. The authors' synchronization procedure depends on the Tracheal microphone, a high-quality $48\,\mathrm{kHz}$ external device, and the Snore microphone, a low-quality $500\,\mathrm{Hz}$ device connected to the PSG machine. The original procedure goes as follows:

- The Tracheal microphone is resampled to $500\,\mathrm{Hz}$ to match the lower quality signal of the other device.
- The Hilbert transform of both signals is derived to robustly assess the envelopes and avoid phase-related disturbances.
- The cross-correlation of the two signals is maximized, finding the lag between the two.

Models trained on these data were found to perform poorly in the regression task: there was a mismatch between the phases of the labels and the predictions. After careful examination, I found much stronger evidence for relative clock drift than for a constant lag. This drift could not be corrected by a simple time translation. Fig. 3.1 shows how the absolute lag changes over the night. Each data point is evaluated as prescribed by the authors on $150\,\mathrm{s}$ windows throughout the night.

Therefore, I decided to model the synchronization as an affine transformation:

$$t_{\mathrm{PSG}} = a\, t_{\mathrm{audio}} + b,$$

where b accounts for the initial offset and a corrects the clock drift. While the exact values of the parameters depend on the patient, they are generally not negligible.

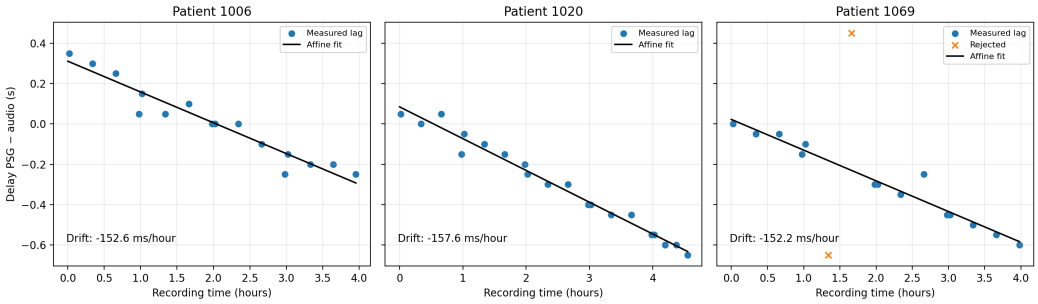


Figure 3.1: Lags evaluated throughout the night are consistent with a constant drift in the clocks.

3.1.2 Time-Frequency Representation

The aligned audio is transformed into a magnitude mel spectrogram. The exact configuration is given by a Hann window of 40 milliseconds, a step of 10 milliseconds, and 128 mel bands covering the available frequency range up to $24\,\mathrm{kHz}$. A 60-second recording therefore produces an image of size 128×6000. The difference between a mel spectrogram and a common spectrogram is that the former squeezes the less relevant higher frequencies using a logarithm. This type of procedure was developed to better represent human perception, so an increase in performance in snore discrimination is not obvious at all.

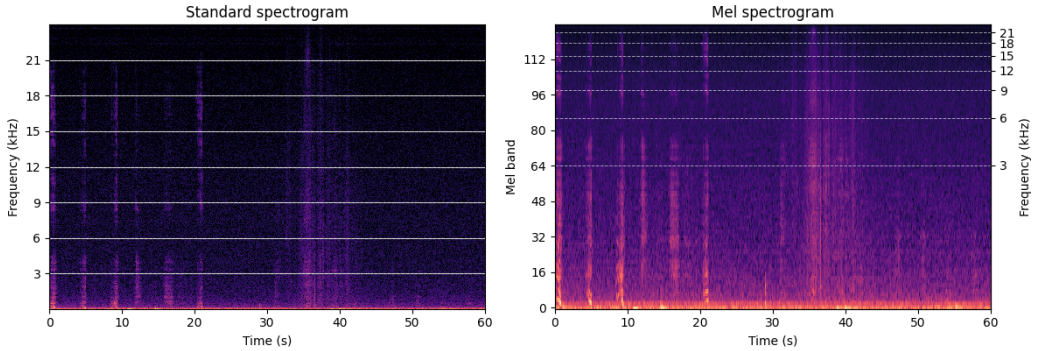


Figure 3.2: A comparison between a standard spectrogram and a mel spectrogram. The segment is from patient 1010 at 1100 seconds. An apnea covers the interval from 21 to 31 seconds, followed by an arousal.

3.1.3 PCEN: Per-Channel Energy Normalization

Domain generalization represents a central difficulty of the project. Thus, any preprocessing that reduces environmental differences in the audio while preserving clinically relevant information should be beneficial. I observed improvements in generalization after applying PCEN (Per-Channel Energy Normalization), a technique first introduced in the field of speech recognition [4] which implements automatic gain control based on dynamic compression.

$$P_{f,t} = \left(\frac{E_{f,t}}{(\epsilon + M_{f,t})^{\alpha}} + \delta \right)^{r} - \delta^{r} \qquad M_{f,t} = (1 - s) M_{f,t-1} + s E_{f,t}.$$

where E is the mel magnitude and M is its exponentially smoothed version. I use $s = 0.025$, $\alpha = 0.98$, $\delta = 2$, $r = 0.5$, and $\epsilon = 10^{-6}$. Smoothing is initialized from the first frame of

each window. As can be seen in Fig. 3.3, this greatly reduces noise and makes snoring more pronounced.

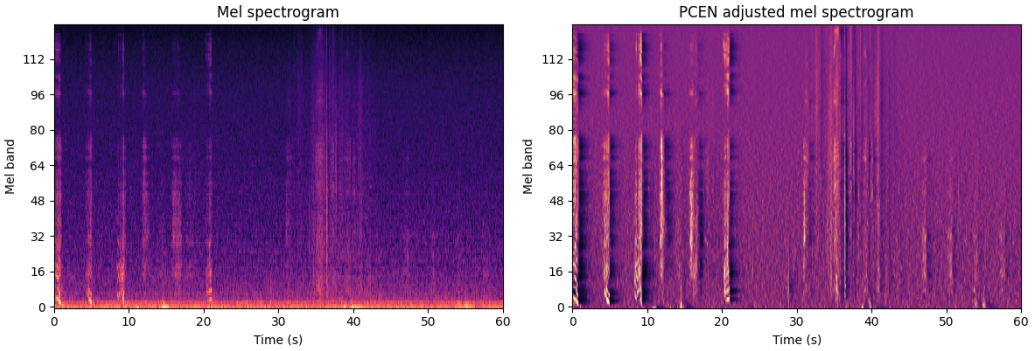


Figure 3.3: A comparison between a mel spectrogram and its PCEN corrected version. The same time window as in Fig. 3.2 is selected.

3.2 Physiological Signal Preprocessing

3.2.1 Filtering and resampling

The four physiological channels are filtered before reducing their sampling rate from $100\,\mathrm{Hz}$ to $10\,\mathrm{Hz}$. The RML annotations identify intervals in which a sensor failed. These annotations are used as validity masks for guiding supervision during regression, while validity is handled separately for each channel. From inspecting the data, it can easily be observed that the validity masks are of low quality: many regions displaying apparent respiratory rhythms are marked as sensor failures, while the converse is also true. I suspect it would be possible to obtain marginally better results by relabeling valid regions with an appropriate algorithm, but this idea has not been tested.

3.2.2 Pressure canonicalization

During some early training runs, I observed that nasal pressure was poorly reconstructed even after perfecting the temporal alignment of the audio, as explained in subsection 3.1.1. I concluded that patients can be split into two cohorts displaying opposite pressure polarity. The idea that a channel can sometimes produce a consistently inverted signal is supported by specific instrument manuals. I avoid asking the model to guess an otherwise unknowable polarity; instead, I assign a consistent orientation to this channel in preprocessing.

The two cohorts are inferred by computing the correlation between pressure and the derivative of abdominal effort in valid 60-second windows. Obtaining a consistent result during the night is necessary and is always observed.

3.2.3 Amplitude normalization

Physiological amplitudes vary considerably across subjects and sensors. A drift can also be observed during the same night, perhaps due to changes in the position of the patient. I normalize each channel using valid samples from a 10-minute context centered on each

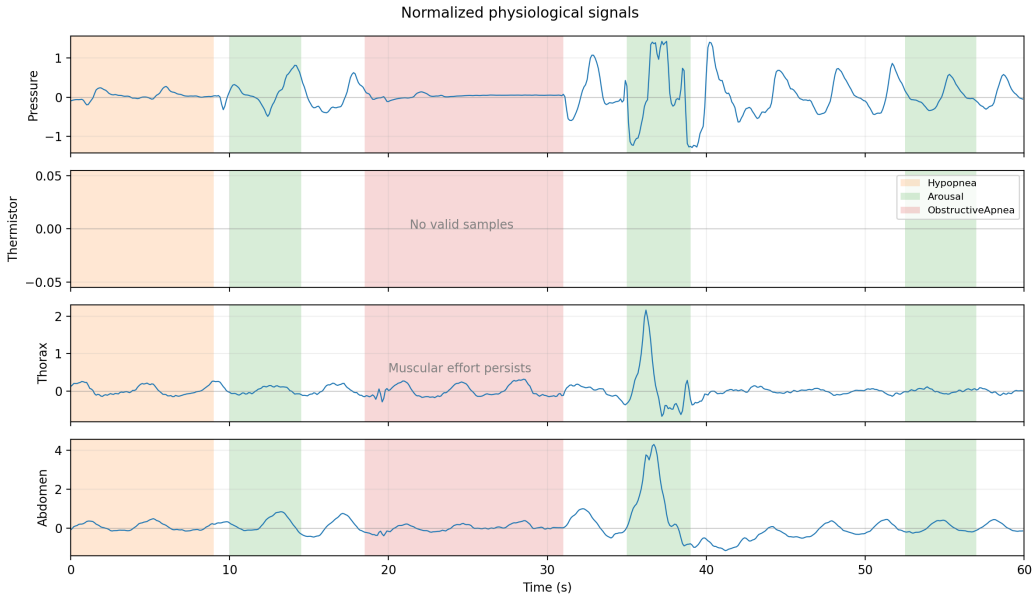


Figure 3.4: Normalized physiological signals in the same time window as in Fig. 3.2 and Fig. 3.3, enriched with RML annotations.

sample. The median provides the local baseline, while the scale is estimated from the fifth and ninety-fifth percentiles:

$$\widetilde{y}_c(t) = \frac{y_c(t) - \mathrm{median}(y_c)}{q_{95,c} - q_{05,c}}.$$

Samples with $|\widetilde{y}_c(t)| \geq 10$ are excluded from the regression loss through the validity mask. This transformation is applied mainly to reduce supervision noise, and to avoid encouraging the model to memorize the identities of the patients, such that more relevant features can be learned.

3.3 Dataset Construction

The final preprocessing pipeline extracts sequential, nonoverlapping 60-second windows from the audio. Each example contains the PCEN image, the four normalized physiological targets at $10\,\mathrm{Hz}$, their validity masks, and a classification label. A window is considered positive if its total overlap with annotated apnea intervals is at least ten seconds. It is negative only if it contains neither apnea nor hypopnea. Windows that do not meet these classification criteria may be used during the reconstruction phase, but are not used for the classification task due to label ambiguity.

I construct three patient-wise splits, using approximately 70% of patients for training, 15% for validation, and 15% for testing. Using multiple splits allows a more robust assessment of the methodology.

Patients are excluded when the audio alignment is missing or insufficiently supported, or when technical problems prevent preprocessing. I also exclude patients with a long final interval without respiratory annotations despite continued desaturation events, as this suggests incomplete scoring. The final cohort contains 261 patients, with 183 used for training, 39 for validation, and 39 for testing in each split. The same patient splits are used for both microphones.

Chapter 4

Methodology

As I briefly explained in the introduction, the classifier learns in two phases. Firstly, I train a backbone with a specialized *regression head* to concurrently reconstruct all four physiological signals. Secondly, I attach a classification head that squeezes the learned features into a single logit, whose sigmoid gives the probability of the given window containing an apnea (according to the convention established in section 3.3).

4.1 Neural Network Architecture

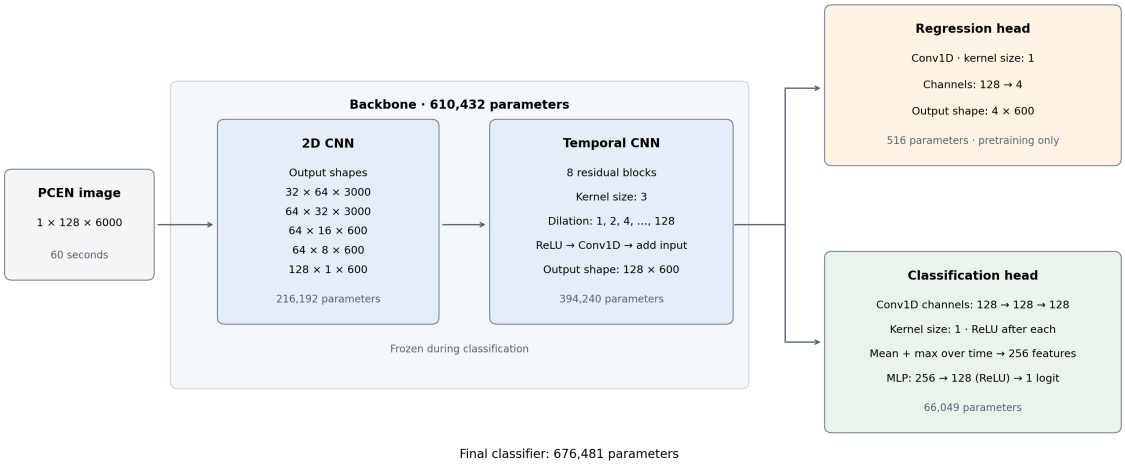


Figure 4.1: Model architecture with shared backbone and task-specific regression and classification heads

The backbone is a standard 2D CNN, augmented with dilations over time on later layers, such that the features of each timestep collect information from the full 60-second window. More precisely, the backbone takes as input a monochromatic image with shape (1, 128, 6000), where 128 is the number of mel bands and $100\,\mathrm{Hz}$ is the temporal frame rate, and produces as output a tensor of shape (128, 600), given that the frequency of the supervision signals is $10\,\mathrm{Hz}$. The *regression head* takes the backbone's temporal features and applies a temporal 1D convolution with unitary kernel size to predict the four physiological channels for each

instant. The *classification head* applies a sequence of convolutions with unitary kernel with a constant number of channels, then aggregates the features across time by concatenating mean and max pooling, and finally applies a shallow MLP to obtain a single logit. The backbone, *regression head*, and *classification head* have a total of 610.432, 516, and 66.049 parameters, respectively. This means that the final model, which can discard the *regression head*, has a total of 676.481 parameters. The exact architecture implementation can be found in the Appendix.

4.2 Regression phase

In the first training phase, the model is guided by a regression objective. I selected the Huber loss in place of the MSE loss because, even though the signals have been appropriately normalized, the ChannelFail masks are imperfect. This choice reduces the influence of outliers and transients in sensor acquisitions. The Huber loss ℓ_δ is defined as

$$\ell_\delta(e) = \begin{cases} \frac{1}{2}e^2 & \text{if } |e| \le \delta \\ \delta\left(|e| - \frac{\delta}{2}\right) & \text{otherwise} \end{cases} \tag{4.1}$$

where e denotes the prediction error and δ controls the transition between quadratic and linear penalties. In this project, δ is set to 1.
The regression loss gives equal weight to each patient and, within each patient, to each available channel:

$$\mathcal{L}_{\text{reg}} = \frac{1}{P}\sum_{p=1}^{P}\frac{1}{|\mathcal{C}_p|}\sum_{c\in\mathcal{C}_p}\bar{\ell}_{p,c}, \tag{4.2}$$

where P is the number of patients, $\mathcal{C}_p$ contains the channels with valid samples for patient p, and $\bar{\ell}_{p,c}$ is the mean Huber loss over all valid samples of that patient's channel, across all windows.
Validation and test losses are computed directly using this patient-balanced average. During training, an equivalent objective is obtained by weighting losses inside the mini-batch. To limit overfitting, I apply early-stopping with a patience of 15 by computing the validation loss every fifth of an epoch. The final model is the one with the lowest validation loss.
Thus, I jointly train the backbone and regression head using AdamW, with a learning rate of 3×10^{-4}, weight decay of 10^{-4}, and a batch size of 32. Gradients are clipped to a maximum norm of 1.

4.3 Classification phase

In the second training phase, I train the classification head attached to the frozen backbone. This head takes the temporal features and produces a single logit for each 60-second window. I use the binary cross-entropy loss:

$$\ell_{\text{BCE}}(z, y) = -y\log\sigma(z) - (1-y)\log(1-\sigma(z)), \tag{4.3}$$

where z is the predicted logit, $y \in \{0, 1\}$ is the target, and σ is the sigmoid function. As in the regression phase, each patient contributes equally to the loss:

$$\mathcal{L}_{\text{cls}} = \frac{1}{P}\sum_{p=1}^{P}\frac{1}{W_p}\sum_{k=1}^{W_p}\ell_{\text{BCE}}(z_{p,k}, y_{p,k}), \tag{4.4}$$

where W_p is the number of included windows belonging to patient p. During training, this objective is implemented through weighted minibatch losses.
I train the classification head using AdamW, with a constant learning rate of 3×10^{-4}, weight decay of 10^{-4}, and a batch size of 64. Gradients are clipped to a maximum norm of 1. Validation is performed every fifth of an epoch, with early stopping after 15 checks. The checkpoint with the lowest patient-balanced validation BCE is retained. Finally, I select a decision threshold by maximizing the mean per-patient balanced accuracy on the validation set. Here, the balanced accuracy is simply the mean of sensitivity and specificity:

$$\mathrm{BA} = \frac{1}{2}\left(\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} + \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}\right) \qquad \mathrm{BA}_{\mathrm{macro}} = \frac{1}{N}\sum_{p=1}^{N} \mathrm{BA}_p. \tag{4.5}$$

Chapter 5

Results

Firstly, I discuss the results pertaining to the signal reconstruction task. Secondly, I show that the classification head learns to use the backbone's temporal features to predict the correct apnea label. While small changes in architecture design have been tested, I only show the runs corresponding to the final selected model.

5.1 Regression phase

During the first training phase, the model learns to reconstruct the four physiological signals based on either the ambient or tracheal sound recordings. Fig. 5.1 makes it apparent that the task is much easier using the tracheal microphone, probably because the signal-to-noise ratio is greater and because there are fewer differences between domains. For the split shown, early stopping triggered after 18 and 14 epochs, respectively, selecting the checkpoints at 15 and 10.2 epochs.

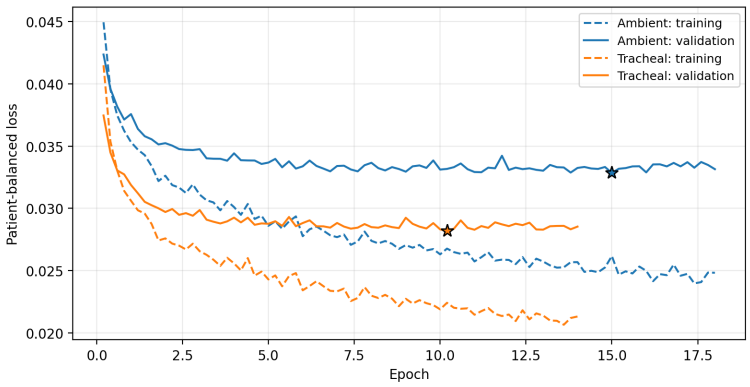


Figure 5.1: Regression loss curves for split 2027. The stars represent the models selected by early stopping.

Mic.	Train	Val.	Test
Ambient	0.026	0.032	0.032
Tracheal	0.023	0.028	0.028

Table 5.1: Patient-balanced loss of the selected regression models, averaged over the three splits.

The regression losses of the selected checkpoints are reported in Tab. 5.1. A more detailed analysis can look at individual patients or channels. Fig. 5.2 shows the distribution of the loss evaluated on each patient separately. The wide range of values demonstrates that some patients are more difficult to predict than others. In particular, many test set patients achieve a lower loss than some training set patients. This remains true when using the Tracheal microphone. We can also look at the per-channel loss distribution. Tab. 5.2 shows that the

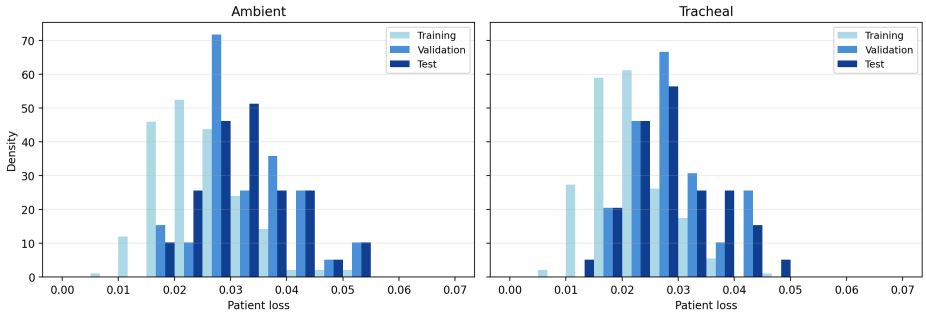


Figure 5.2: Per-patient loss distributions for split 2027.

Tracheal microphone improves regression loss over each considered channel. Moreover, they agree on the order of increasing reconstruction complexity: thermistor, pressure, abdomen and thorax. This could be due to the fact that the information needed to precisely guess a specific channel may be partial in the audio, and some channels are less determined than others.

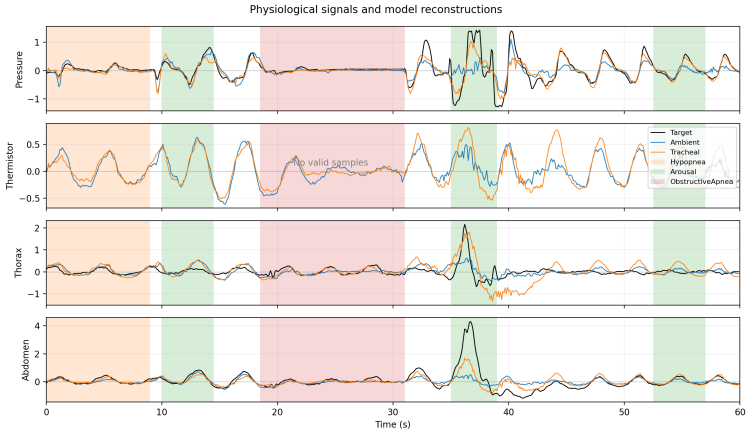


Figure 5.3: Reconstructed signals in the same time window as in Fig. 3.2, Fig. 3.3, and Fig. 3.4. The Tracheal microphone supports a better reconstruction.

Mic.	Pressure	Thermistor	Thorax	Abdomen
Ambient	0.025	0.023	0.046	0.034
Tracheal	0.022	0.017	0.043	0.031

Table 5.2: Patient-balanced test loss for each physiological channel, averaged over the three splits.

I also inspected some reconstruction failures in the training, validation, and test sets. Some predictions show phase errors, in the worst cases resembling polarity inversions. For nasal pressure, there was evidence of inconsistent polarity in the recordings themselves, which was corrected during preprocessing (see Sec. 3.2.2). However, the remaining failures cannot be explained as easily, since they are not consistent throughout the same night. I think that better polishing the data and understanding these reconstruction breakdowns could further improve both model performance and the features learned during pretraining.

5.2 Classification phase

In this section, I compare two training schemes: the first trains a classification head attached to the frozen regression backbone to make use of its rich features; the second trains the full model end-to-end with a randomly initialized backbone. In such a way, we can assess whether we have successfully used the regression signal to boost the classification performance.

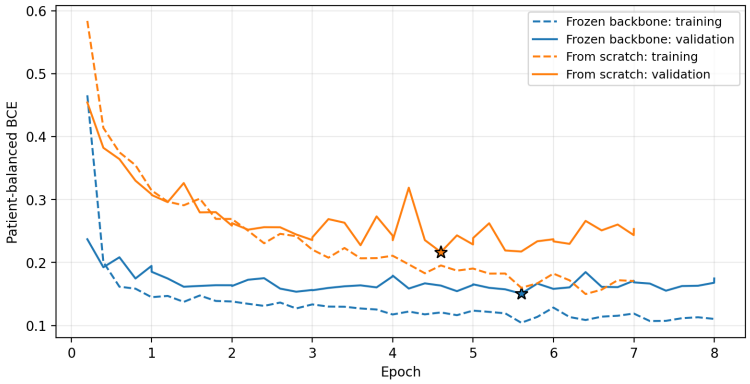


Figure 5.4: Classification loss curves (Tracheal audio) for split 2027. The stars represent the models selected by early stopping.

Model	Train	Val.	Test
Pretrained backbone	0.111	0.146	0.155
From scratch	0.133	0.186	0.197

Table 5.3: Patient-balanced BCE of the selected classification models using tracheal audio, averaged over the three splits.

Fig. 5.4 shows the progression of the training and validation loss using the tracheal audio (equivalent results are obtained using the Ambient microphone). While the faster convergence and head start of the regression-pretrained model were expected, the picture also shows that it plateaus at a lower BCE loss. The advantage is confirmed by looking at more relevant classification metrics. Tab. 5.4 shows the relevant metrics robustly averaged over three different splits of the dataset. More details are shown in Tab. 5.5.

Model	BA (%)	Macro BA (%)	Sens. (%)	Spec. (%)	AP (%)	AUROC (%)
From scratch (Ambient)	89.17	85.39	89.99	88.34	97.78	96.09
Pretrained (Ambient)	91.79	88.02	91.21	92.37	98.39	97.15
From scratch (Tracheal)	92.64	88.71	91.52	93.77	98.63	97.67
Pretrained (Tracheal)	93.86	91.00	95.36	92.36	99.20	98.61

Table 5.4: Classification metrics show an advantage of the regressively pretrained model.

Model	Split A	Split B	Split C
From scratch (Ambient)	88.95	88.96	89.59
Pretrained (Ambient)	91.29	91.68	92.40
From scratch (Tracheal)	92.77	91.52	93.64
Pretrained (Tracheal)	94.43	92.89	94.25

Table 5.5: Test balanced accuracy (%) for each patient-wise split.

We can make a few observations:

- Pretraining improves performance on all three splits. For ambient audio, balanced accuracy increases by 2.34, 2.72, and 2.81 percentage points, respectively. For tracheal audio, the increases are 1.66, 1.38, and 0.60 points. The advantage is therefore consistent, although smaller for the tracheal microphone.
- Tracheal audio gives better results than ambient audio, both with and without pretraining. Even the tracheal model trained from scratch performs better, on average, than the pretrained ambient model.
- The improvement also holds when averaging balanced accuracy across patients: pretraining adds 2.63 points for ambient audio and 2.29 points for tracheal audio.

Chapter 6

Conclusions

With this project, I investigated whether reconstructing physiological signals from breathing sounds could help a model learn useful features for apnea detection. In particular, regression pretraining followed by a frozen-backbone classifier outperformed end-to-end classification from random initialization. Averaged over the three splits, test balanced accuracy increased from 92.64% to 93.86% for tracheal audio, and from 89.17% to 91.79% for ambient audio. While some doubts remain regarding the identification of all four channels based on audio alone, these results support the usefulness of physiological supervision during training.
Moreover, model performance varied depending on the recording modality during regression and classification alike. This highlights the central role of audio quality and standardization in apnea detection when viewed as a domain generalization problem.
A limitation that makes it difficult to compare the performance of my models versus previous attempts in the literature is that different labeling procedures are often used. For example, Professor Quadrini's work constructs the dataset in a different fashion, using 30-second segments for negative windows and variable-length windows for positive windows.
Future work could investigate whether physiological pretraining also improves the distinction between obstructive, central, and mixed apneas as a multiclass classification objective, as these events differ in their airflow and respiratory effort patterns which I attempted to reconstruct.

Appendix A

Model implementation

```
import torch
from torch import nn


class Backbone(nn.Module):
    def __init__(self):
        super().__init__()

        self.cnn = nn.Sequential(
            nn.Conv2d(1, 32, kernel_size=(5, 5), stride=(2, 2), padding=(2, 2)),  # (B, 32, 64, 3000)
            nn.ReLU(),

            nn.Conv2d(32, 64, kernel_size=(5, 5), stride=(2, 1), padding=(2, 2)),  # (B, 64, 32, 3000)
            nn.ReLU(),

            nn.Conv2d(64, 64, kernel_size=(3, 5), stride=(2, 5), padding=(1, 2)),  # (B, 64, 16, 600)
            nn.ReLU(),

            nn.Conv2d(64, 64, kernel_size=(3, 3), stride=(2, 1), padding=(1, 1)),  # (B, 64, 8, 600)
            nn.ReLU(),

            nn.Conv2d(64, 128, kernel_size=(8, 1)),  # (B, 128, 1, 600)
        )

        self.tcn = nn.ModuleList([
            nn.Sequential(
                nn.ReLU(),
                nn.Conv1d(128, 128, kernel_size=3, dilation=d, padding=d),
            )
            for d in (1, 2, 4, 8, 16, 32, 64, 128)
        ])

    def forward(self, x):
        x = self.cnn(x).squeeze(2) # (B, 128, 600)
        for block in self.tcn:
            x = x + block(x)
        return x


class RegressionHead(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv = nn.Conv1d(128, 4, kernel_size=1)  # (B, 4, 600)

    def forward(self, x):
        return self.conv(x)  # (B, 4, 600)


class ClassificationHead(nn.Module):
    def __init__(self):
        super().__init__()
```

```
        self.conv = nn.Sequential(
            nn.Conv1d(128, 128, kernel_size=1),
            nn.ReLU(),
            nn.Conv1d(128, 128, kernel_size=1),
            nn.ReLU(),
            )


        self.mlp = nn.Sequential(
            nn.Linear(256, 128),
            nn.ReLU(),
            nn.Linear(128, 1),
        )

    def forward(self, x):
        x = self.conv(x)  # (B, 128, 600)
        x = torch.cat([x.mean(-1), x.amax(-1)], dim=1)
        return self.mlp(x)  # (B, 1), raw logit


class Model(nn.Module):
    def __init__(self):
        super().__init__()
        self.backbone = Backbone()
        self.regression_head = RegressionHead()
        self.classification_head = ClassificationHead()

    def forward(self, x):
        features = self.backbone(x)
        signals = self.regression_head(features)
        logit = self.classification_head(features)
        return signals, logit
```

Bibliography

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